Abstract Algebra

SF1678

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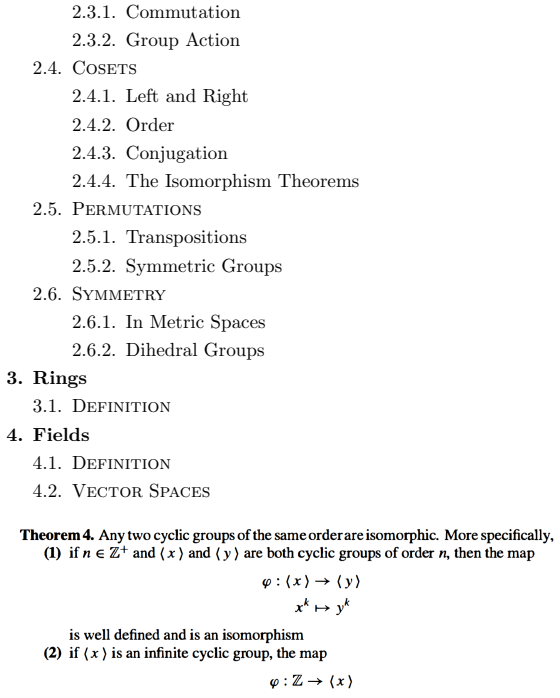


Theorem 4. Any two cyclic groups of the same order are isomorphic. More specifically,
(1) if $n \in \mathbb{Z}^+$ and $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n, then the map

$$\varphi : \langle x \rangle \to \langle y \rangle$$
$$x^k \mapsto y^k$$

is well defined and is an isomorphism
(2) if $\langle x \rangle$ is an infinite cyclic group, the map

$$\varphi : \mathbb{Z} \to \langle x \rangle$$
$$k \mapsto x^k$$

is well defined and is an isomorphism.

Proof: Suppose $\langle x \rangle$ and $\langle y \rangle$ are both cyclic groups of order n. Let $\varphi : \langle x \rangle \to \langle y \rangle$ be defined by $\varphi(x^k) = y^k$; we must first prove φ is well defined, that is,

$$\text{if } x^r = x^s, \text{ then } \varphi(x^r) = \varphi(x^s).$$

Since $x^{r-s} = 1$, Proposition 3 implies $n \mid r - s$. Write $r = tn + s$ so

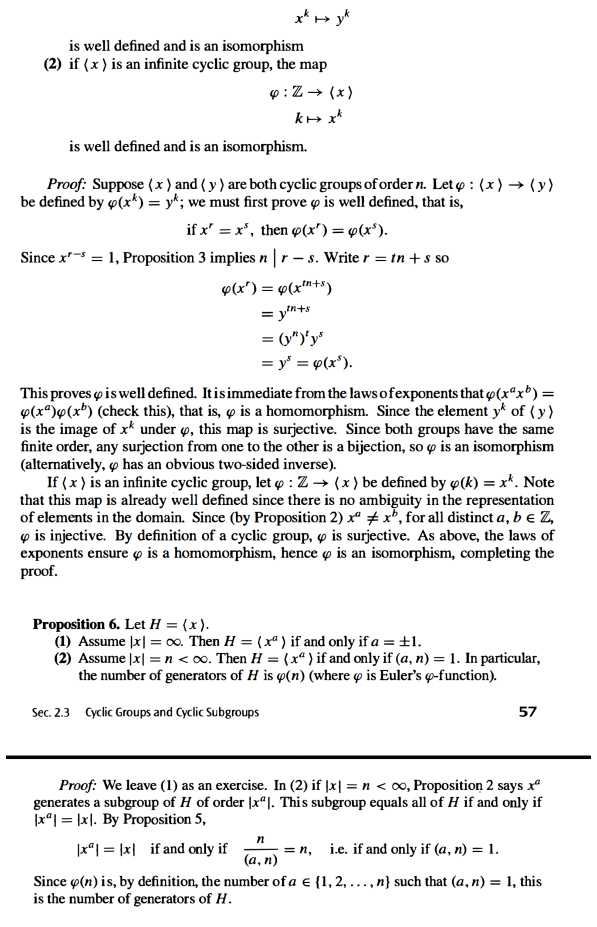
$$\begin{aligned}\varphi(x^r) &= \varphi(x^{tn+s}) \\ &= y^{tn+s} \\ &= (y^n)^t y^s \\ &= y^s = \varphi(x^s).\end{aligned}$$

This proves φ is well defined. It is immediate from the laws of exponents that $\varphi(x^a x^b) = \varphi(x^a)\varphi(x^b)$ (check this), that is, φ is a homomorphism. Since the element y^k of $\langle y \rangle$ is the image of x^k under φ, this map is surjective. Since both groups have the same finite order, any surjection from one to the other is a bijection, so φ is an isomorphism (alternatively, φ has an obvious two-sided inverse).

If $\langle x \rangle$ is an infinite cyclic group, let $\varphi : \mathbb{Z} \to \langle x \rangle$ be defined by $\varphi(k) = x^k$. Note that this map is already well defined since there is no ambiguity in the representation of elements in the domain. Since (by Proposition 2) $x^a \neq x^b$, for all distinct $a, b \in \mathbb{Z}$, φ is injective. By definition of a cyclic group, φ is surjective. As above, the laws of exponents ensure φ is a homomorphism, hence φ is an isomorphism, completing the proof.

Proposition 6. Let $H = \langle x \rangle$.
(1) Assume $|x| = \infty$. Then $H = \langle x^a \rangle$ if and only if $a = \pm 1$.
(2) Assume $|x| = n < \infty$. Then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$. In particular, the number of generators of H is $\varphi(n)$ (where φ is Euler's φ-function).

Sec. 2.3 Cyclic Groups and Cyclic Subgroups 57



Proof: We leave (1) as an exercise. In (2) if $|x| = n < \infty$, Proposition 2 says x^a generates a subgroup of H of order $|x^a|$. This subgroup equals all of H if and only if $|x^a| = |x|$. By Proposition 5,

$$|x^a| = |x| \quad \text{if and only if} \quad \frac{n}{(a,n)} = n, \quad \text{i.e. if and only if } (a, n) = 1.$$

Since $\varphi(n)$ is, by definition, the number of $a \in \{1, 2, \dots, n\}$ such that $(a, n) = 1$, this is the number of generators of H.

Theorem 7. Let $H = \langle x \rangle$ be a cyclic group.
(1) Every subgroup of H is cyclic. More precisely, if $K \leq H$, then either $K = \{1\}$ or $K = \langle x^d \rangle$, where d is the smallest positive integer such that $x^d \in K$.
(2) If $|H| = \infty$, then for any distinct nonnegative integers a and b, $\langle x^a \rangle \neq \langle x^b \rangle$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, where $|m|$ denotes the absolute value of m, so that the nontrivial subgroups of H correspond bijectively with the integers $1, 2, 3, \dots$.
(3) If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a. This subgroup is the cyclic group $\langle x^d \rangle$, where $d = \frac{n}{a}$. Furthermore, for every integer m, $\langle x^m \rangle = \langle x^{(n,m)} \rangle$, so that the subgroups of H correspond bijectively with the positive divisors of n.

Proof: (1) Let $K \leq H$. If $K = \{1\}$, the proposition is true for this subgroup, so we assume $K \neq \{1\}$. Thus there exists some $a \neq 0$ such that $x^a \in K$. If $a < 0$ then since K is a group also $x^{-a} = (x^a)^{-1} \in K$. Hence K always contains some positive power of x. Let

$$\mathcal{P} = \{b \mid b \in \mathbb{Z}^+ \text{ and } x^b \in K\}.$$

By the above, $\mathcal{P}$ is a nonempty set of positive integers. By the Well Ordering Principle (Section 0.2) $\mathcal{P}$ has a minimum element — call it d. Since K is a subgroup and $x^d \in K$, $\langle x^d \rangle \leq K$. Since K is a subgroup of H, any element of K is of the form x^a for some integer a. By the Division Algorithm write

$$a = qd + r \qquad 0 \leq r < d.$$

Then $x^r = x^{(a-qd)} = x^a (x^d)^{-q}$ is an element of K since both x^a and x^d are elements of K. By the minimality of d it follows that $r = 0$, i.e., $a = qd$ and so $x^a = (x^d)^q \in \langle x^d \rangle$. This gives the reverse containment $K \leq \langle x^d \rangle$ which proves (1).

We leave the proof of (2) as an exercise (the reasoning is similar to and easier than the proof of (3) which follows).

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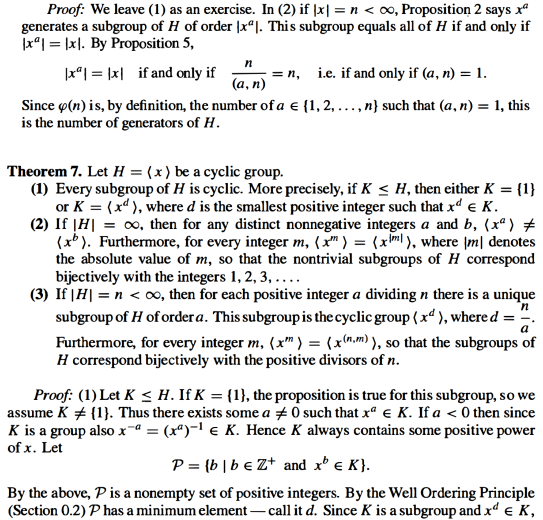


(3) Assume $|H| = n < \infty$ and $a \mid n$. Let $d = \frac{n}{a}$ and apply Proposition 5(3) to obtain that $\langle x^d \rangle$ is a subgroup of order a, showing the existence of a subgroup of order a. To show uniqueness, suppose K is any subgroup of H of order a. By part (1) we have

$$K = \langle x^b \rangle$$

where b is the smallest positive integer such that $x^b \in K$. By Proposition 5

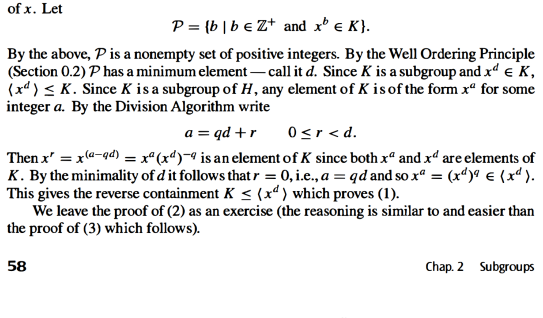
$$\frac{n}{d} = a = |K| = |x^b| = \frac{n}{(n, b)},$$

so $d = (n, b)$. In particular, $d \mid b$. Since b is a multiple of d, $x^b \in \langle x^d \rangle$, hence

$$K = \langle x^b \rangle \leq \langle x^d \rangle.$$

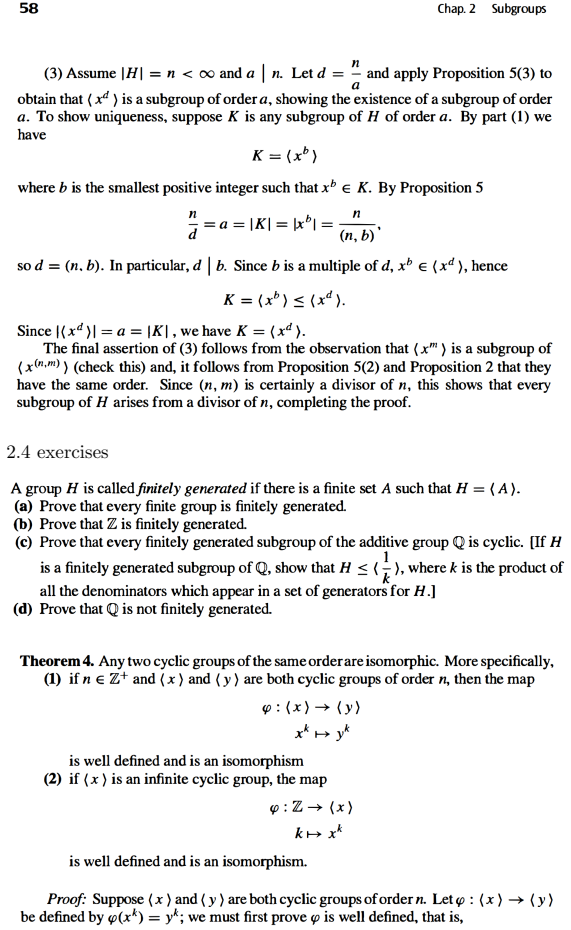
Since $|\langle x^d \rangle| = a = |K|$, we have $K = \langle x^d \rangle$.

The final assertion of (3) follows from the observation that $\langle x^m \rangle$ is a subgroup of $\langle x^{(n,m)} \rangle$ (check this!), and it follows from Proposition 5(2) and Proposition 2 that they have the same order. Since (n, m) is certainly a divisor of n, this shows that every subgroup of H arises from a divisor of n, completing the proof.



2.4 exercises

A group H is called *finitely generated* if there is a finite set A such that $H = \langle A \rangle$.
(a) Prove that every finite group is finitely generated.
(b) Prove that $\mathbb{Z}$ is finitely generated.
(c) Prove that every finitely generated subgroup of the additive group $\mathbb{Q}$ is cyclic. [If H is a finitely generated subgroup of $\mathbb{Q}$, show that $H \leq \langle \frac{1}{k} \rangle$, where k is the product of all the denominators which appear in a set of generators for H.]
(d) Prove that $\mathbb{Q}$ is not finitely generated.



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find a cyclic infinite group

find an infinite subset that is closed under operation but not inverse

Theorem 1

If p is the smallest prime index of G, then all subgroups of order p are normal.

Theorem 2 (Cauchy's or Prime Divisors)

If p is a prime that divides $|G|$, then G must have a subgroup *and an element* of order p.

Theorem 3 (Sylow's or Prime Power Divisors)

If p^k is prime *power* and divides $|G|$, then G must have a subgroup of order p^k.

Definition 1 (p-group)

Let G be a group, let p be a prime, let α be a positive integer, and let m be a positive integer such that $m \nmid \alpha$.

(i) If $|G| = p^\alpha$ for some $\alpha \geq 1$, then G is a **p-group** and its subgroups are **p-subgroups**.
(ii) If $|G| = p^\alpha m$, then the $n_p(G)$ subgroups $\mathrm{Syl}_p(G)$ of order p^α are **Sylow p-subgroups**.

Lemma

Let $P \in \mathrm{Syl}_p(G)$. If Q is a p-subgroup of G, then

$$Q \cap N_G(P) = Q \cap P.$$

Theorem 4 (Sylow's)

Let G be a group of order $p^\alpha m$.

(i) $\mathrm{Syl}_p(G) \neq \varnothing$
(ii) If P is a Sylow p-subgroup and Q is a p-subgroup, then

$$\exists g \in G : Q \leq gPg^{-1},$$

i.e., Q is contained in some conjugate of P. In particular, any two Sylow p-subgroups of G are conjugate in G.
(iii) $n_p \equiv 1 \pmod{p}$.
(iv) $n_p = [G : N_G(P)]$ for every $P \in \mathrm{Syl}_p(G)$.

Corollary

Let $P \leq G$ be a Sylow p-subgroup. Then the following are equivalent:

(i) P is the only Sylow p-subgroup of G, i.e., $n_p(G) = 1$
(ii) P is normal in G
(iii) P is characteristic in G
(iv) All subgroups generated by elements of p-power order are p-groups, i.e., if X is any subset of G such that $|x|$ is a power of p for all $x \in X$, then $\langle X \rangle$ is a p-group.

1. Sets

1.1. RELATIONS

1.1.1. Set Theory

Definition 1.1 (Tuple)

A tuple (or sometimes **list**) is a finite, ordered collection of elements, possibly with repetitions.

An n-tuple is of length n. A 2-tuple is called an **ordered pair**.

1.1.2. Morphisms

Definition 1.2 (Map)

Subsets of the Cartesian product $A \times B$ are called **relations**. A **map** is a **well-defined** relation written

$$f \subseteq A \times B \quad \text{or} \quad f : A \to B \quad \text{or} \quad A \xrightarrow{f} B$$

and is

- **surjective** or **onto** if $f(A) = B$, which is equivalent to being **right invertible** with $f \circ f^{-1} = \mathrm{id}_B$.
- **injective** or **one-to-one** if $a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$, which is equivalent to being **left invertible** with $f^{-1} \circ f = \mathrm{id}_A$.
- **bijective** or **a one-to-one correspondence** if both above apply, which is equivalent to being **invertible**.

Definition 1.3 (Homomorphism)

A **homomorphism** is a map between algebraic structures that preserves the structure.

A group homomorphism $\varphi : G \to H$ satisfies

$$\varphi(g * \gamma) = \varphi(g) \cdot \varphi(\gamma)$$

for all $g, \gamma \in G$.

Theorem 1.1

If $\varphi : G \to H$ is a group homomorphism,

(i) $\varphi(\mathrm{id}_G) = \mathrm{id}_H$
(ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ for all $g \in G$
(iii) $\varphi(G) \leq H$

Definition 1.4 (Isomorphism)

An **isomorphism** is a bijective homomorphism.

If there exists an isomorphism between two algebraic structures, they are said to be **isomorphic**, denoted with ≅.

Definition 1.5 (Endomorphism)

An **endomorphism** is a homomorphism from a mathematical object to itself.

Definition 1.6 (Automorphism)

An **automorphism** is simultaneously an isomorphism and an endomorphism: a bijective homomorphism from an object to itself.

Example. The **identity map** id_X is a trivial automorphism defined

$$\mathrm{id}_X(x) = x \qquad \forall x \in X.$$

Example. An automorphism of a set is called a **permutation**.

1.1.3. Equivalence

Definition 1.7 (Equivalence Relation)

An **equivalence relation** on a set A is a relation $R \subseteq A \times A$ with the following properties:

(i) **reflexive** $(a, a) \in R \qquad \forall a \in A$
(ii) **symmetric** $(a, b) \in R \Longrightarrow (b, a) \in R$
(iii) **transitive** $(a, b), (b, c) \in R \Longrightarrow (a, c) \in R$

If $(x, y) \in R$ we write xRy or $x \sim y$.

Let R be an equivalence relation on a set A.

Definition 1.8 (Equivalence Class)

The **equivalence class** of an element $a \in A$

$$[a] = \bar{a} = \{b \in A \mid b \sim a\}$$

is the set of all elements related to it.

Example. The modulo m congruence class $[a] = \{a + km \mid k \in \mathbb{Z}\}$.

Definition 1.9 (Quotient Set)

The **quotient set** of A by R

$$A/R = \{[a] \mid a \in A\}$$

is the set of all equivalence classes induced by the relation.

Definition 1.10 (Partition)

A **partition** $\mathcal{P}$ of a set A is a collection of non-empty, disjoint subsets of A whose union is A.

Theorem 1.2

The quotient set A/R forms a partition of A.

Definition 1.11 (Quotient Map)

A **quotient map** is a map $\pi : A \to A/R$ defined by

$$\pi(a) = [a].$$

Theorem 1.3

Quotient maps are surjective.

1.2. ℤ

1.2.1. Linear Diophantine Equations

Let a and b be integers with $\gcd(a, b) = d$.

Lemma (Bézout's Identity)

There exist integers x and y such that $ax + by = d$.

Lemma (Euclid's)

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Theorem 1.4 (Solvability Condition)

The linear Diophantine equation $ax + by = c$ has (infinitely many) solutions if and only if $d \mid c$.

Theorem 1.5 (Solutions)

The solutions, if they exist, are

$$x = x_0 + t\frac{b}{d}, \quad y = y_0 - t\frac{a}{d}, \quad \text{for } t \in \mathbb{Z}.$$

Theorem 1.6 (Euclidean Algorithm)

Let a and b be non-negative integers with $a \geq b$. Then there exist unique integers q and r such that

$$a = qb + r, \quad 0 \leq r < b.$$

Example. We apply the algorithm to compute $\gcd(252, 105)$:

$$\begin{aligned}252 &= 2 \cdot 105 + 42 \\ 105 &= 2 \cdot 42 + 21 \\ 42 &= 2 \cdot 21 + 0\end{aligned}$$

Hence, $\gcd(252, 105) = 21$.

Example. We then use our results to solve $252x + 105y = 21$.

Since $\gcd(252, 105) = 21$, we now work backwards:

$$\begin{aligned}21 &= 105 - 2 \cdot 42 \\ &= 105 - 2 \cdot (252 - 2 \cdot 105) \\ &= 105 - 2 \cdot 252 + 4 \cdot 105 \\ &= 5 \cdot 105 - 2 \cdot 252 \\ &= -2 \cdot 252 + 5 \cdot 105\end{aligned}$$

So a particular solution is $x_0 = -2, y_0 = 5$. The general solutions are

$$x = -2 + 5t, \quad y = 5 - 12t, \quad \text{for } t \in \mathbb{Z}.$$

2. Groups

2.1. DEFINITIONS

2.1.1. Groups

Definition 2.1 (Group)

A **group** $(G, *)$ is a nonempty set G with a binary operation

$$* : G \times G \to G$$

such that for all $a, b, c \in G$, it satisfies

(i) (Closure) $a * b \in G$
(ii) (Associativity) $(a * b) * c = a * (b * c)$
(iii) (Identity) $\exists e \in G : e * a = a * e = a$
(iv) (Inverse) $\exists a^{-1} \in G : a * a^{-1} = a^{-1} * a = e$

Additionally, a group is called **abelian** or **commutative** if it is

(v) (Commutative) $a * b = b * a$

Examples. The **general linear group** $GL_n(\mathbb{R})$ consists of all matrices in $\mathbb{R}^{n \times n}$ with $\det \neq 0$. The **special linear group** $SL_n(\mathbb{R})$ additionally requires that $\det = 1$.

Tip

The proof of (iii) and (iv) can be shortened if (v) is proven.

Note

Defying conventional wisdom, **abelian** is usually not capitalized.

2.1.2. Subgroups

Let $(G, *)$ be a group.

Definition 2.2 (Subgroup)

A subset $H \subseteq G$ is a **subgroup** if it is itself a group under the operation inherited from G.

Theorem 2.1 (Subgroup Criterion)

A subset $H \subseteq G$ is a subgroup if and only if

(i) $H \neq \varnothing$ (nonempty)

and

(ii) $a * b^{-1} \in H \quad \forall a, b \in H$

or

(ii) $a * b \in H \quad \forall a, b \in H$
(iii) $a^{-1} \in H \quad \forall a \in H$

or

(ii) $a * b \in H \quad \forall a, b \in H$ (closed)
(iii) $|H| < \infty$ (finite)

2.1.3. Cyclic

Definition 2.3 (Cyclic Subgroup)

For every $a \in G$, the set

$$\langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$$

forms a **cyclic subgroup** of G.

Theorem 2.2

The cyclic subgroup $\langle a \rangle \leq G$ is the smallest subgroup containing a.

Definition 2.4 (Generating Set)

The subgroup $\langle S \rangle = \langle s, s^{-1} \mid s \in S \rangle = \{s^k \mid s \in S, k \in \mathbb{N}\}$ consists of all *finite* products of elements in its **generating set** S and their inverses.

Definition 2.5 (Generator)

An element $a \in G$ is a **generator** of G if $\langle a \rangle = G$.

Definition 2.6 (Cyclic Group)

We say G is a **cyclic group** if it contains a **generator** of itself.

Theorem 2.3

All subgroups of a cyclic group are cyclic.

Theorem 2.4

Every cyclic group is abelian.

Proof. Let g and γ be elements of a cyclic group $G = \langle a \rangle$. Then there exist integers m and n such that $g = a^m$ and $y = a^n$, so

$$g\gamma = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = \gamma g.$$

2.2. $\mathbb{Z}_n$

Let a and n be positive integers and let p be prime.

Definition 2.7 ($\mathbb{Z}_n$)

The **integers modulo n** is a set of congruence classes

$$\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$$

that form an abelian group under $+$ and a monoid under $\times$, together forming a ring with a group of units called the **multiplicative group of integers modulo n**, denoted $U(n)$ and consisting of the integers coprime to n, with inverse elements defined to satisfy $aa^{-1} (\equiv a^{-1}a) \equiv 1 \pmod{n}$.

Definition 2.8 (Euler's Totient Function)

$$\phi(n) = |U(n)| = |\{k \in \mathbb{Z} \mid 1 \leq k \leq n \text{ and } \gcd(n, k) = 1\}|$$

Example. $\phi(12) = |\{1, 5, 7, 11\}| = 4$

Theorem 2.5 (Euler's)

If a and n are coprime, then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof. The order of a is $|U(n)| = \phi(n)$.

Lemma

$\phi(p) = p - 1$

Theorem 2.6 (Fermat's Little)

If a and p are coprime, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

In general, even if they are not coprime,

$$a^p \equiv a \pmod{p}.$$

2.3. FAMILIES

2.3.1. Commutation

Let X be any nonempty subset of G.

Definition 2.9 (Centralizer)

The **centralizer** of X in G is the set

$$C_G(X) = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in X\},$$

all elements in G that commute with all elements in X, fixing every element of X under conjugation.

Remark

$gxg^{-1} = x \Longleftrightarrow gx = xg$

Note

If X is a singleton set $\{x\}$, we simply write $C_G(x)$.

Definition 2.10 (Center)

The **center** of G is the set

$$Z(G) = C_G(G) = \{g \in G \mid g\gamma g^{-1} = \gamma \text{ for all } \gamma \in G\},$$

all elements in G that commute with all elements in G, fixing every element of G under conjugation.

Definition 2.11 (Normalizer)

The **normalizer** of X in G is the set

$$N_G(X) = \{g \in G \mid gXg^{-1} = X\},$$

all elements in G that commute with X as a set, fixing X as a set under conjugation.

Theorem 2.7

$Z(G) \leq C_G(X) \leq N_G(X) \leq G$, with all equalities if G is abelian.

Theorem 2.8

If X is a subgroup, not just a subset, then $X \leq N_G(X) \leq G$.

Definition 2.12 (Commutator)

Let G be a group with elements $x, y \in G$ and nonempty subsets $A, B \subseteq G$.

(i) The **commutator** of x and y is $[x, y] = x^{-1}y^{-1}xy$.
(ii) $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$
(iii) Define $G' = \langle [x, y] \mid x, y \in G \rangle$, the subgroup of G generated by commutators of elements from G, called the **commutator subgroup** of G.

Theorem 2.9

$[x, y] = \mathrm{id}_G \Longleftrightarrow x$ and y commute

2.3.2. Group Action

Definition 2.13 (Group Action)

A map $\cdot : G \times X \to X$ is a **(left) group action** if it satisfies:

(i) (Identity) $e \cdot x = x$ for all $x \in X$.
(ii) (Compatibility) $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

Let $\cdot : G \times X \to X$ be a group action.

Definition 2.14 (Orbit)

The **orbit** of $x \in X$ is the set

$$G \cdot x = \{g \cdot x \mid g \in G\},$$

all points in X that x can be moved to by the group action.

Definition 2.15 (Transitive)

The action of G on X is **transitive** if there is only one orbit, i.e., given any pair $x, y \in X$ there is some $g \in G$ such that $g \cdot x = y$.

Theorem 2.10

The orbits of a cyclic subgroup $\langle \sigma \rangle \leq S_n$ is the sets of numbers in the distinct cycles given by its cycle decomposition.

Example. The orbits of $\langle (1\ 2)(3\ 4\ 5) \rangle$ are $\{1, 2\}$ and $\{3, 4, 5\}$.

Definition 2.16 (Stabilizer)

The **stabilizer** of $x \in X$ is the set

$$G_x = \{g \in G \| g \cdot x = x\},$$

all elements in G that fix x under the group action.

Theorem 2.11 (Orbit–Stabilizer)

$|G \cdot x| = [G : G_x] = |G|/|G_x|$

Definition 2.17 (Fixed Point Set)

The **fixed point set** of $g \in G$ is the set

$$X^g = \{x \in X \mid g \cdot x = x\},$$

all elements in X that are fixed under g.

Definition 2.18 (Kernel)

The **kernel** is the set

$$\ker(\cdot) = \{g \in G \mid g \cdot x = x \text{ for all } x \in X\},$$

all elements in G that fix every element of X under the group action.

Definition 2.19 (Faithful)

A group action is **faithful** if $\ker(\cdot) = \{e\}$.

2.4. COSETS

2.4.1. Left and Right

Let H be a subgroup of a group G.

Definition 2.20 (Coset)

For any $g \in G$, the **left coset** of H in G is

$$gH = \{g * h \mid h \in H\}.$$

Similarly, the **right coset** is

$$Hg = \{h * g \mid h \in H\}.$$

An element of a coset is a **representative** for that coset.

Theorem 2.12

The left and right cosets respectively partition G.

Theorem 2.13

There are equally many left and right cosets of H in G.

2.4.2. Order

Definition 2.21 (Order)

The **order of a group** $|G|$ is the number of elements in G.

The **order of an element** $|g|$ is the smallest positive integer n such that $g^n = e$. If there is no such n, the order is infinite.

Theorem 2.14 (Classification)

If G is a cyclic group of order n, then

(i) $G \cong \mathbb{Z} \iff n = \infty$
(ii) $G \cong \mathbb{Z}_n \iff n < \infty$

Definition 2.22 (Direct Product)

The **direct product** $G_1 \times G_2 \times \dots \times G_n$ of the sets $G_1, G_2, \dots, G_n$ with the operations $*_1, *_2, \dots, *_n$ is the set of n-tuples with operation defined componentwise:

$$(g_1, \dots, g_n) * (\gamma_1, \dots, \gamma_n) = (g_1 *_1 \gamma_1, \dots, g_n *_n \gamma_n).$$

If there are infinitely many groups, the n-tuples are replaced by sequences.

Theorem 2.15

$$G/G_i \cong G_1 \times \dots \times G_{i-1} \times G_{i+1} \times \dots \times G_n$$

Theorem 2.16

$$|G_1 \times G_2 \times \dots \times G_n| = |G_1|\ |G_2| \cdots |G_n|$$

Theorem 2.17

Let $x_i = (\mathrm{id}_1, \dots, \mathrm{id}_{i-1}, g_i, \mathrm{id}_{i+1}, \dots, n)$ for some $g_i \in G_i$. Then

$$|x_1 x_2 \cdots x_n| = \mathrm{lcm}(|x_1|, |x_2|, \dots, |x_n|).$$

Lemma

Let

Theorem 2.18

Let $m, n \in \mathbb{Z}^+$. Then $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \Longleftrightarrow \gcd(m, n) = 1$.

Proof. We must prove both implications.

Consider the element $g = (1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$ which has order $|g| = \mathrm{lcm}(|1_m|, |1_n|) = \mathrm{lcm}(m, n)$.

First assume $\gcd(m, n) = 1$. Then $|g| = mn$

Definition 2.23 (Bounded Group)

$\exists N \in \mathbb{Z}^+ : g^N = e \ \forall g \in G$

Definition 2.24 (Torsion Group)

$\forall g \in G\ \exists n \in \mathbb{Z}^+ : g^n = e$

Examples. Finite groups. Bounded groups.

Definition 2.25 (Torsion Subgroup)

If G is abelian, its torsion subgroup is an abelian subgroup

$$T(G) = \{g \in G : \exists n \in \mathbb{Z}^+ : g^n = e\} = \{g \in G : |g| < \infty\}.$$

Two special cases arise:

(i) If $T(G) = G$ then G is a **torsion abelian group**.
(ii) If $T(G) = \{e\}$ then G is a **torsion-free abelian group**.

Definition 2.26 (Index)

The **index** of H in G is the number of left cosets, denoted $[G : H]$.

Theorem 2.19 (Lagrange's)

Let G be finite. Then the order of H divides the order of G.

Corollary

If G is of prime order, it

(i) lacks proper nontrivial subgroups
(ii) is generated by any element $g \neq e$
(iii) is cyclic

Theorem 2.20 (Lagrange's, Generalized)

Regardless of whether G is finite or infinite,

$$|G| = [G : H]\ |H|.$$

Proof. G has $|G|$ elements and is partitioned into $[G : H]$ distinct left cosets, each of which has $|H|$ elements.

Corollary

If $H \trianglelefteq G$ then $[G : H] = |G|/|H| = |G/H|$.

Theorem 2.21

Let $(g, h) \in G \times H$ be a tuple of groups. If they have finite orders $|g| = r < \infty$ and $|h| = s < \infty$, then $|(g, h)| = \mathrm{lcm}(r, s)$.

Corollary

$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1$

2.4.3. Conjugation

Definition 2.27 (Conjugation)

The **conjugation** by $a \in G$ is an **inner automorphism** of G defined $g \mapsto gag^{-1}$.

Definition 2.28 (Normal Subgroup)

The following are equivalent for a subgroup H of G:

(i) H is a **normal subgroup** of G, denoted $H \trianglelefteq G$.
(ii) $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.
(iii) $gHg^{-1} \subseteq H$ for all $g \in G$.
(iv) $gHg^{-1} = H$ for all $g \in G$.
(v) $gH = Hg$ (equal left coset and right coset) for all $g \in G$.
(vi) The sets of left and right cosets coincide.

Examples. Trivially, $G \trianglelefteq G$ and $\{e\} \trianglelefteq G$.

Warning

Normalness is not transitive, i.e.,

$$A \trianglelefteq B \trianglelefteq C \not\Rightarrow A \trianglelefteq C.$$

Note

Some authors use $\triangleleft$ exclusively for proper normal subgroups, while others use it for all normal subgroups.

The same principle holds for the subset sign $\subset$.

Theorem 2.22

Every subgroup of index 2 is normal.

Proof. Let $H \leq G$ be the subgroup such that $[G : H] = 2$. We wish to show that $gH = Hg$ for all $g \in G$.

If $g \in H$, then $gH = H = Hg$. Otherwise, if $g \notin H$, then $gH = G/H = Hg$ since $G = H \cup Hg = G = H \cup gH$.

Thus, $gH = Hg$ for all g, so H is a normal subgroup of G.

Theorem 2.23

If $H \leq G$ and $K \leq G$, then $HK \leq G$ if and only if $HK = KH$.

Corollary

If $gH = H\gamma$ for some $g, \gamma \in G$, then $g = \gamma$ and $g \in N_G(H)$.

Theorem 2.24

All subgroups of abelian groups are normal subgroups.

Definition 2.29 (Simple Group)

A group is **simple** if it has no normal nontrivial proper subgroup.

Definition 2.30 (Fiber)

Let $\varphi : G \to \Omega$ be a surjective map from a group to a set. The **fiber** of φ over some $\omega \in \Omega$ is its preimage in G

$$\varphi^{-1}(y) = \{g \in G \mid \phi(g) = \omega\},$$

the set of all elements in the domain that map to ω.

Let $\varphi : G \to H$ be a group homomorphism.

Definition 2.31 (Fiber of Group)

The **fiber** of φ over some $h \in H$ is

$$\varphi^{-1}(h) = \{g \in G \mid \varphi(g) = h\}.$$

Theorem 2.25

The collection of nonempty fibers forms a partition of G. Hence, every $g \in G$ is only in the fiber over $\varphi(g)$.

Definition 2.32 (Kernel)

The **kernel** of a group homomorphism φ is the set

$$\ker \varphi = \varphi^{-1}(\mathrm{id}_H) = \{g \in G \mid \varphi(g) = \mathrm{id}_H\}.$$

Let $K = \ker \varphi$.

Theorem 2.26

$K \trianglelefteq G$

Theorem 2.27

For every $g \in G$ such that $\varphi(g) = h$ the entire fiber is

$$\varphi^{-1}(h) = gK = Kg.$$

In words, the fibers of a homomorphism are the cosets of its kernel.

Definition 2.33 (Quotient Group, Alternative)

The quotient group is

$$G/K = \{\varphi^{-1}(h) \mid h \in \varphi(G)\},$$

the set of all nonempty fibers of φ.

If φ is surjective, then $\varphi(G) = H$ and all fibers are nonempty.

Definition 2.34 (Quotient Group)

Let $N \trianglelefteq G$. The **quotient group** or **factor group** is

$$G/N = \{gN \mid g \in G\} (= \{Ng \mid g \in G\}),$$

the set of all left cosets of N in G, paired with the group operation

$$(aN) * (bN) = (ab)N \quad \forall a, b \in G,$$

which is well-defined if and only if $N \trianglelefteq G$.

Example.

name	symbol	set	operation
large group	G	$\mathbb{Z}$	$+$
normal subgroup	N	$n\mathbb{Z}$	$+$
quotient group	G/N	$\mathbb{Z}_n$	$+ \bmod n$

Definition 2.35 (Natural Projection)

Let $N \trianglelefteq G$. The natural/canonical projection/homomorphism is the map $\pi : G \to G/N$ defined by $\pi(g) = gN$ for all $g \in G$.

Theorem 2.28

For π defined above:

(i) π is a homomorphism
(ii) π is surjective, i.e., $\pi(G) = G/N$
(iii) $\ker \pi = N$

2.4.4. The Isomorphism Theorems

Let $\varphi : G \to H$ be a group homomorphism.

Theorem 2.29 (First or Fundamental of Homomorphisms)

(i) $\ker \varphi \trianglelefteq G$
(ii) $\varphi(G) \cong G/\ker \varphi$
(iii) $\varphi(G) \leq H$

Corollary

(i) $|\varphi(G)| = [G : \ker \varphi]$

The following are equivalent:

(ii) φ is surjective
(iii) $H \cong G/\ker \varphi$
(iv) $|G| = [G : \ker \varphi]$ *Proof.* See (i).

The following are equivalent:

(v) φ is injective
(vi) $\ker \varphi = \mathrm{id}_G$

Lemma

If A and B are subgroups of G, then

$$AB \leq G \Leftrightarrow AB = BA$$

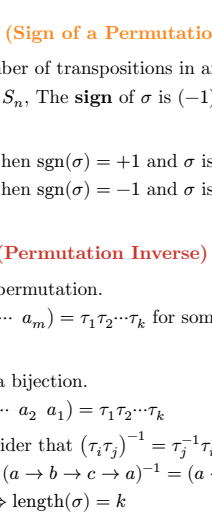
Furthermore, if A and B are finite, then

$$|AB| = \frac{|A||B|}{|A \cap B|}.$$

Theorem 2.30 (Second or Diamond or Parallelogram)

Let $S \leq G$ and $N \leq G$ be subgroups such that $S \leq N_G(N)$. Then,

(i) $SN \leq G$ *Proof.* See the above lemma.
(ii) $N \trianglelefteq SN$
(iii) $S \cap N \trianglelefteq S$
(iv) $(SN)/N \cong S/(S \cap N)$



2.5. PERMUTATIONS

2.5.1. Transpositions

Definition 2.36 (Transposition)

A **transposition** of the elements $i, j \in \{1, \dots, n\}$ is the permutation τ_{ij} with the cycle notation $(i\ j)$.

Theorem 2.31

Any permutation can be written as compositions of transpositions.

Theorem 2.32

The identity permutation may only be written as an even number of transpositions.

Theorem 2.33 (Order)

$|\sigma| = \mathrm{length}(\sigma)$

Definition 2.37 (Sign of a Permutation)

Let k be the number of transpositions in any decomposition of a $(k+1)$-cycle $\sigma \in S_n$. The **sign** of σ is $(-1)^k$.

Hence,

(i) if k is **even** then $\mathrm{sgn}(\sigma) = +1$ and σ is said to be even.
(ii) if k is **odd** then $\mathrm{sgn}(\sigma) = -1$ and σ is said to be odd.

Theorem 2.34 (Permutation Inverse)

Let $\sigma \in S_n$ be a permutation.

(i) $\sigma = (a_1\ a_2\ \cdots\ a_m) = \tau_1 \tau_2 \cdots \tau_k$ for some $m \leq n$ and k
(ii) σ^{-1} exists

Proof. σ is a bijection.

(iii) $\sigma^{-1} = (a_m\ \cdots\ a_2\ a_1) = \tau_1 \tau_2 \cdots \tau_k$

Proof. Consider that $(\tau_i \tau_j)^{-1} = \tau_j^{-1} \tau_i^{-1} = \tau_j \tau_i$ and $(a\ b\ c)^{-1} = (a \to b \to c \to a)^{-1} = (a \leftarrow b \leftarrow c \leftarrow a)$.

(iv) $\sigma^{-1} = \sigma^k \Longleftrightarrow \mathrm{length}(\sigma) = k$

Proof. The order of a k-cycle is k.

(v) $\mathrm{sgn}(\sigma^{-1}) = \mathrm{sgn}(\sigma) = (-1)^k$

Definition 2.38 (Fixed Point))

A **fixed point** is invariant under a transformation such as a permutation.

2.5.2. Symmetric Groups

Definition 2.39 (Symmetric Group)

The **symmetric group** of the finite set $X = \{1, 2, \dots, n\}$ is

$$S_n = \{\sigma : X \to X \mid \sigma \text{ is bijective}\}$$

and consists of $n!$ permutations, which form **the symmetric group on n letters** under the operation of composition

Theorem 2.35

The automorphism group of a set is its symmetric group.

Definition 2.40 (Permutation Group)

A subgroup of a symmetric group S_n is a **permutation group**.

Definition 2.41 (Alternating Group)

The **alternating group**

$$A_n = \{\sigma \in S_n \mid \mathrm{sgn}(\sigma) = 1\}$$

is the subgroup of S_n consisting of all even permutations.

It has $n!/2$ elements for $n \geq 2$.

Theorem 2.36

The number of k-cycles is

$$\binom{n}{k}(k-1)! = \frac{n!}{(n-k)!k!}(k-1)! = \frac{n!}{(n-k)!k}$$

in S_n. In A_n there are equally many if k is odd, but 0 if k is even.

Problem. Show that A_4 has no subgroup of order 6.

Solution. Assume the opposite; suppose there is a subgroup $H \leq A_4$ such that $|H| = 6$.

By Lagrange's theorem, $[A_4 : H] = |A_4|/|H| = 12/6 = 2$. Theorem 2.22 dictates that any subgroup of index 2 is normal, so $H \triangleleft A_4$. Since the quotient group has order $|A_4/H| = [A_4 : H] = 2$,

$$(gH)^2 = H \implies g^2 H = H \implies g^2 \in H$$

for all $g \in A_4$.

There are 8 distinct 3-cycles in A_4. The inverse of any 3-cycle $g \in A_4$ is $g^{-1} = g^2 \in H$, since the length is 3. Hence, $g \in H$. However, there are 8 distinct 3-cycles in A_4, so it follows that $|H| \geq 8 > 6$.

Theorem 2.37

The sign map $\mathrm{sgn} : S_n \to \{1, -1\}$ is a group homomorphism.

The kernel of this homomorphism is the alternating group

$$\ker(\mathrm{sgn}) = A_n.$$

2.6. SYMMETRY

2.6.1. In Metric Spaces

Definition 2.42 (Isometry)

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is an **isometry** if it perserves distances, i.e.,

$$d_X(x_1, x_2) = d_Y(f(x_1), f(x_2)) \quad \forall x_1, x_2 \in X.$$

Definition 2.43 (Symmetry)

A **symmetry** of an object $O \subseteq X$ in a metric space (X, d) is an isometry $f : X \to X$ with the image $f(O) = O$.

Informally, a symmetry of a geometric object is a rearrangment of the object preserving the arrangement of its sides and vertices as well as its distances and angles.

Examples. Some types of symmetries:

(i) **Reflectional** Reflection across a hyperplane (a line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$)
(ii) **Rotational** Rotation within a plane ($\mathbb{R}^2$ is rotated around a fixed point, while $\mathbb{R}^3$ is rotated around an axis).
(iii) **Translational** All points are shifted by a constant vector.
(iv) **Identity** No change.

Definition 2.44 (Rigid Motion)

A **rigid motion** is a map $f : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$\|u - v\| = \|f(u) - f(v)\| \quad \forall u, v \in \mathbb{R}^n.$$

Equivalently, it is a surjective isometry between Euclidean spaces

Example. A symmetry of an object in an Euclidean space $X = \mathbb{R}^n$ is also a rigid motion.

2.6.2. Dihedral Groups

Definition 2.45 (Dihedral Group)

The **dihedral group** D_n or D_{2n} is the group of symmetries (or equivalently rigid motions) of an n-gon. Its degree is n and its order is $2n$.

Theorem 2.38

The dihedral group can be represented as

$$D_n = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle.$$

3. Rings

3.1. DEFINITION

Definition 3.1 (Ring)

A **ring** is a set R with two binary operations $+$ and $\cdot$ such that

(i) $(R, +)$ is an abelian group with identity 0.
(ii) $(R, \cdot)$ is a monoid with identity 1.
(iii) Multiplication distributes over addition:
for all $a, b, c \in R$, $a \cdot (b \cdot c) = a \cdot b + a \cdot c$.

Definition 3.2 (Group of Units)

Let $(R, +, \cdot)$ be a ring. The **group of units** $U(R)$ is the set of all elements in R that have multiplicative inverses, i.e.,

$$U(R) = \{a \in R \mid \exists a^{-1} \in R : a \cdot a^{-1} = a^{-1} \cdot a = 1\}.$$

4. Fields

4.1. DEFINITION

Definition 4.1 (Field)

A **field** is a set F with two binary operations $+$ and $\cdot$ such that

(i) $(F, +)$ is an abelian group with identity 0.
(ii) $(F \setminus \{0\}, \cdot)$ is an abelian group with identity 1.
(iii) Multiplication distributes over addition:
for all $a, b, c \in F$, $a \cdot (b \cdot c) = a \cdot b + a \cdot c$.

Examples. $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{R}(x), \mathbb{Q}(i)$, and $\mathbb{F}_p$ for prime p.

4.2. VECTOR SPACES

Definition 4.2 (Vector Space)

A vector space over a field F is a set V with two operations:

- Vector addition $+ : V \times V \to V$
- Scalar multiplication $\cdot : F \times V \to V$

These must satisfy

(i) (Closure under addition) $\mathbf{u} + \mathbf{v} \in V$
(ii) (Distributivity over vectors) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
(iii) (Associativity) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
(iv) (Identity) $\mathbf{u} + \mathbf{0} = \mathbf{u}$

(v) (Inverse) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
(vi) (Commutativity) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

(vii) (Closure under scalar multiplication) $a\mathbf{u} \in V$
(viii) (Distributivity over scalars) $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$
(ix) (Associativity) $a(b\mathbf{u}) = (ab)\mathbf{u}$
(x) (Identity) $1\mathbf{u} = \mathbf{u}$

Note that requiring closure is often omitted.